

Group 4
Chair of Air Transport Technology and Logistics

Research Task in Air Transport and Logistics

Winter Semester 2025/26
09 December 2025



Table of Content

1. Provided Data
2. Data Cleaning & Feature Extraction
3. Definition of different phase of flight
4. Phase Segmentation
5. Physics Based Parameter and Equation
6. Efficiency and Evaluation Metrics



Research Topic & Motivation

Research Topic:

Energy consumption modeling for UAVs using flight data and mission parameters

Why This Research?

- Help to understand how speed, altitude, wind affect energy use
- Safe Return-to-Home(RTH) decision
- Support mission feasibility and endurance planning



Provided Data

Given UAV Flight Data includes:

- Flight number
- Date and time
- Predefined Altitude
- Position (x,y,z directions)
- Acceleration(x,y,z directions)
- Orientation (x,y,z,w direction)
- Battery status(voltage, current)
- Position (x,y,z directions)
- Wind speed and angle
- Payload

In-flight positional and energy use data set of a DJI Matrice 100 quadcopter for small package delivery

Thiago A. Rodrigues¹, Jay Patrikar², Arnav Choudhry¹, Jacob Feldgoise³, Vaibhav Arcot⁴, Aradhana Gahlaut¹, Sophia Lau⁵, Brady Moon², Bastian Wagner⁶, H. Scott Matthews¹, Sebastian Scherer² & Constantine Samaras¹

We autonomously directed a small quadcopter package delivery Uncrewed Aerial Vehicle (UAV) or "drone" to take off, fly a specified route, and land for a total of 209 flights while varying a set of operational parameters. The vehicle was equipped with onboard sensors, including GPS, IMU, voltage and current sensors, and an ultrasonic anemometer, to collect high-resolution data on the inertial states, wind speed, and power consumption. Operational parameters, such as commanded ground speed, payload, and cruise altitude, were varied for each flight. This large data set has a total flight time of 10 hours and 45 minutes and was collected from April to October of 2019 covering a total distance of approximately 65 kilometers. The data collected were validated by comparing flights with similar operational parameters. We believe these data will be of great interest to the research and industrial communities, who can use the data to improve UAV designs, safety, and energy efficiency, as well as advance the physical understanding of in-flight operations for package delivery drones.

Data Cleaning & Feature Extraction

Data cleaning Steps:

- Converted date + time to unified timestamps
- Removed duplicates and sorted by flight and time
- Applied physical limits to remove impossible values
 - Wind Speed > 60m/s
 - Battery voltage <5V> 30V
 - Velocity > 50 m/s or speed > 100m/s
 - Altitude outside -100 to 50000m

Sanity check Results: Already clean data set and ready to use.

Extracted Features:

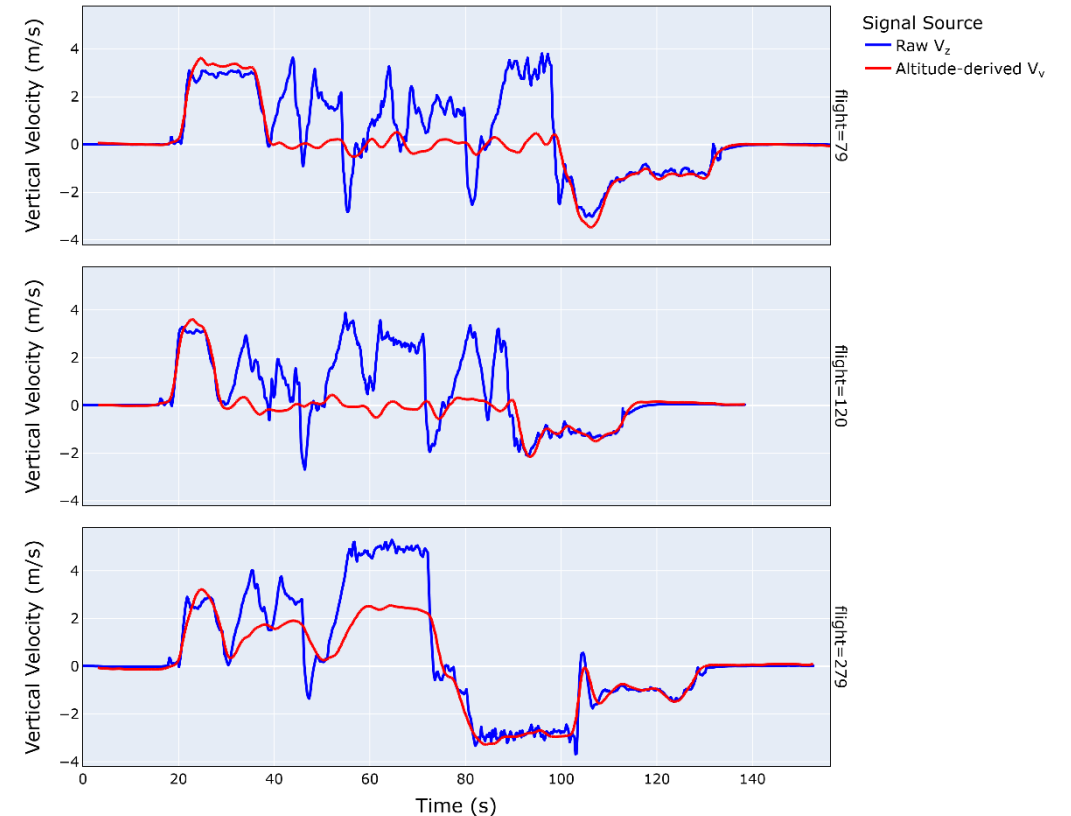
- Horizontal Speed: $v_h = \sqrt{v_x^2 + v_y^2}$
- Altitude Measured: $h = p_z^t - p_z^0$
- Vertical Speed: $v_v = \frac{dh}{dt}$
- Power: $P_{real} = V_b * I_b$

Use of vertical speed v_v from altitude instead of raw v_z

Reason:

- Raw velocity v_z is noisy and unstable, especially in cruise phase.
- Noise in v_z leads to incorrect phase segmentation.
- v_z from altitude gives a physics-consistent vertical speed profile

Raw Vertical Velocity V_z vs Altitude-derived Vertical Velocity V_v



Definition of different phase of flight

Taxi:

Almost no horizontal and vertical speed

Power: $P_{\text{raw}} < 5 \text{ W}$

Hover:

Nearly zero motion

Horizontal speed: $|v_h| < 0.3 \text{ ms}^{-1}$

Vertical speed from altitude: $|v_v| < 0.2 \text{ ms}^{-1}$

Cruise:

Vertical speed : $-0.6 \leq v_v \leq 0.6 \text{ ms}^{-1}$

Altitude: above 80% of max altitude of the flight

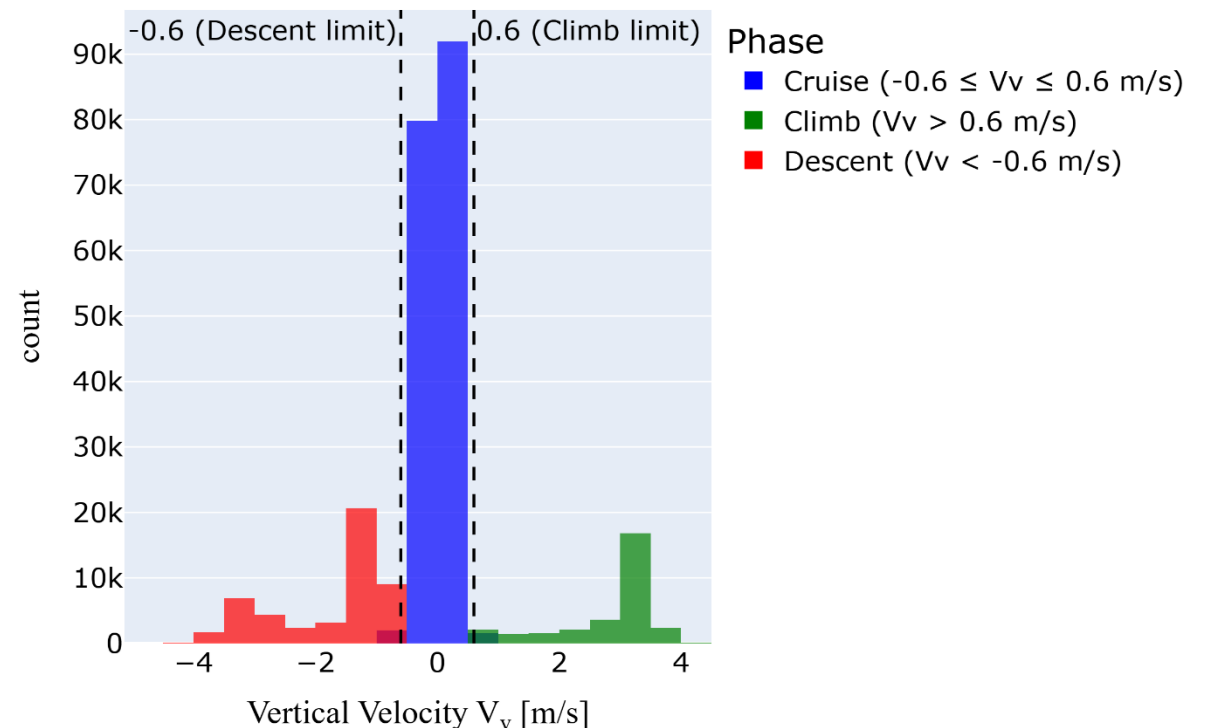
Climb:

Vertical speed: $v_v > 0.6 \text{ ms}^{-1}$

Decent:

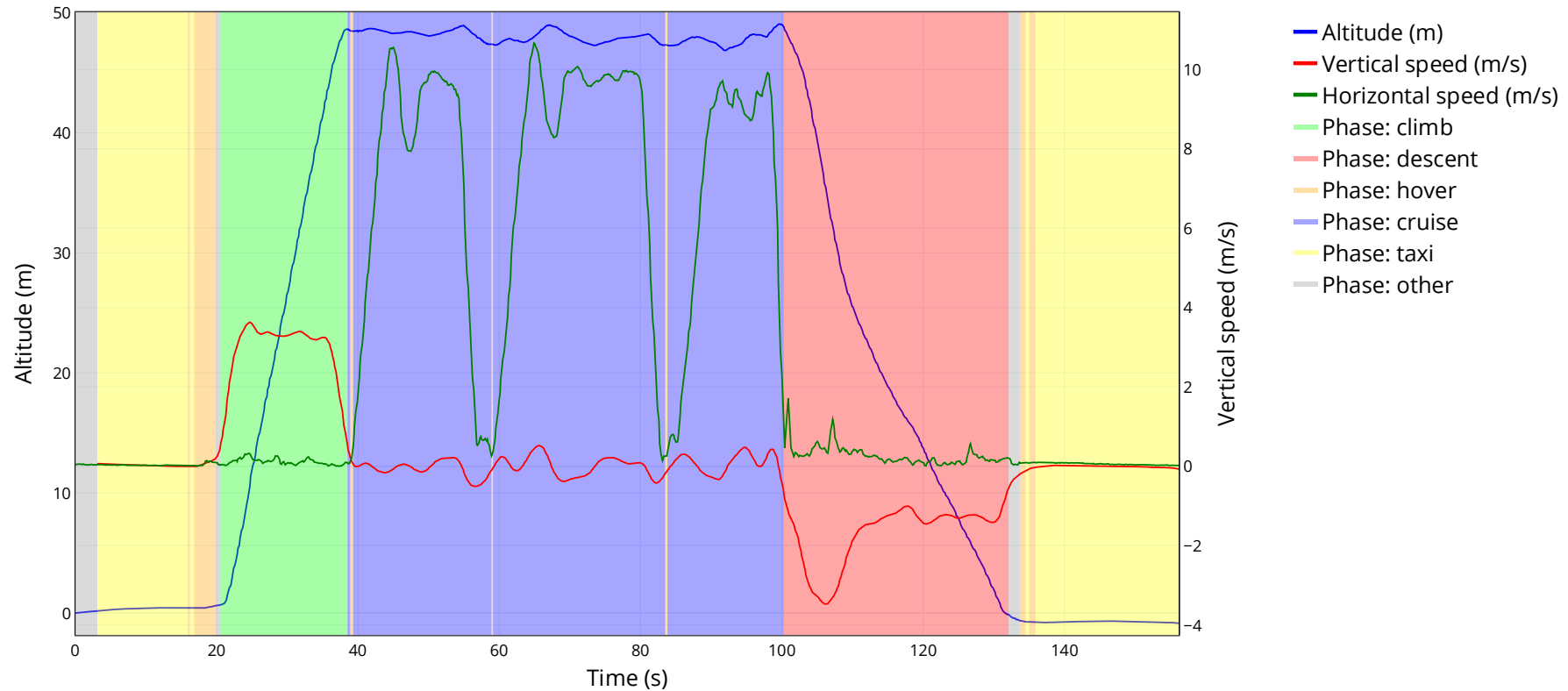
Vertical speed: $v_v < -0.6 \text{ ms}^{-1}$

Vertical Speed Histogram by Phase



Phase Segmentation

Flight 79 – Altitude, Vertical Speed & Phases



Physics based power model development

Parameter	Symbol	Value
Gravity	g	9.81 ms^{-2}
Air Density	ρ	1.225 kgm^{-3}
Empty mass	m_o	3.68 kg
Payload	p	Data set
Total mass	M	$m_0 + p$
Weight	W	$m.g$
Rotar radius	R_{prop}	0165 m
Disk area of one rotor	A_{rot}	0.085 m^2
Total rotor area	A_{tot}	0.34 m^2

Parameter	Symbol	Value
Drag coefficient	C_d	0.15
Reference frontal area	S_{ref}	0.05 m^2
Rotor axis distance	RAD	0.43m
Wind speed	v_{wind}	Data set
Wind angle	θ	Data set
Wind in x-direction	w_x	$-v_{\text{wind}} \cdot \cos(\theta)$
Wind in y direction	w_y	$-v_{\text{wind}} \sin(\theta)$

Physics equations for different phases

Hover Induced Power:

$$P_{hover} = \frac{W^{1.5}}{\sqrt{2\rho A_{tot}}}$$

Climb Induced Power:

$$P_{climb} = P_{hover} \left(\frac{v_v}{2v_{hover}} + \sqrt{1 + \left(\frac{v_v}{2v_{hover}} \right)^2} \right)$$

$$v_{hover} = \sqrt{\frac{W}{2\rho A_{tot}}}$$

Cruise Induced Power:

$$P_{cruise} = P_{hover} + \frac{1}{2} \rho c_d S_{ref} V_{air}^3$$

$$v_{air} = \sqrt{(v_x - w_x)^2 + (v_y - w_y)^2}$$

Decent Induced Power:

$$P_{decent} = P_{hover} \left(-\frac{v_v}{2v_{hover}} + \sqrt{1 + \left(\frac{v_v}{2v_{hover}} \right)^2} \right)$$

$$v_{hover} = \sqrt{\frac{W}{2\rho A_{tot}}}$$

Phase	Sample	Mean Power (w)	Min Power (w)	Max Power (w)
Hover	12722	252.07	236.92	312.92
Climb	29954	321.79	248.01	405.35
Decent	48272	231.11	170.79	300.04
Cruise	103112	300.52	236.92	773.26

Efficiency and evaluation Metrics

Efficiency: How much mechanical power converts to electrical battery power.

$$P_{battery} = \frac{P_{mech}}{\eta}$$

Estimate efficiency per segment using data:

$$\eta = \frac{\sum P_{model}^2}{\sum (P_{real} P_{model})}$$

Segment	η
Hover	0.829
Climb	0.534
Cruise	0.462
Descent	0.611

Mean Absolute Error(MAE):

- Shows how far model power from real power

Root Mean square Error(RMSE):

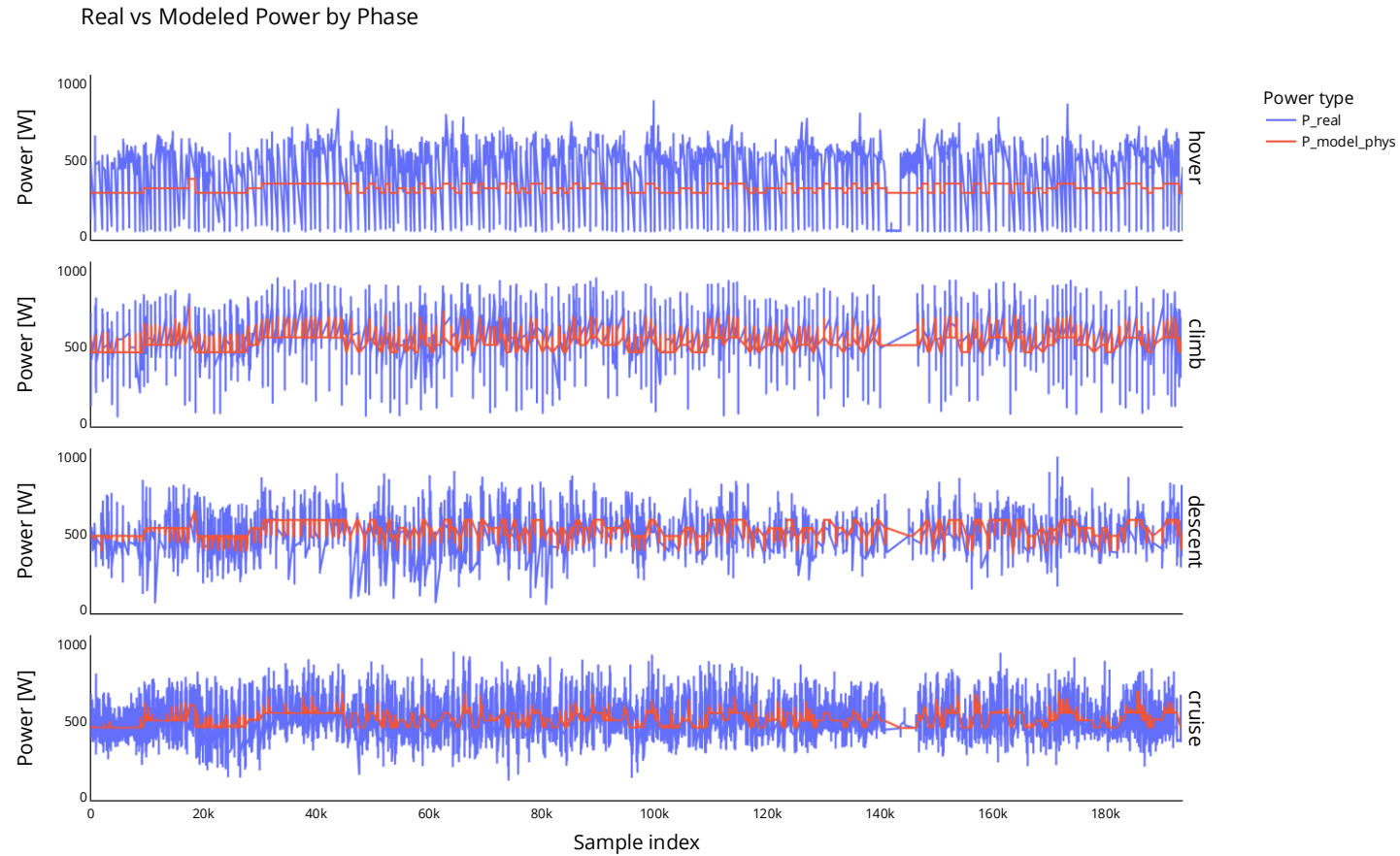
- Shows how big error between model and real power

Mean Absolute Percentage Error:

- Show how accurate is the model

Phase	MAE (w)	RMSE (w)	MAPE (%)
Hover	201.86	211.64	231.04
Climb	57.04	82.64	10.49
Cruise	66.98	91.75	13.42
Descent	45.77	64.36	9.60
All Phase	64.84	97.20	26.02

Real Power vs Model Power



Thank You

